

Pulse-programmed mechanical memory for zero-standby robotic matter

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Robotic matter becomes more powerful when a small unit can change shape, remember that shape without static power, and reset on command. FluxCell frames that problem as pulse-programmed mechanical memory: a Sarrus-compatible unit cell in which a fixed electropermanent-magnetic cartridge changes the overlap of moving steel keepers, coupling magnetic state to visible geometry. The current contribution is a locked architecture, figure-backed design record, and falsifiable one-axis proof protocol. It does not yet claim force, efficiency, lifetime, or reliable two-axis operation; those claims are reserved for measured stroke, current, heat, hold, release, and cycle data.

1 Introduction

Soft robots and robotic materials increasingly distribute function through body geometry, material state, and embedded physical computation rather than through a single remote actuator (1–4). This direction is attractive because mechanical structure can store information, route signals, and shape motion before software has to intervene. A central unresolved question is not simply how to actuate a soft or compliant structure, but how to make a small repeated unit that changes state, stores that state physically, and tiles into a larger machine without paying continuous electrical power to remember its last command.

Electropermanent magnets offer a direct route to low-standby-power state retention. An electropermanent magnet combines magnetic materials with different coercivities so that a short current pulse can switch the net external field while the magnetized state persists after current returns to zero (5–7). EPs have been used for programmable matter, modular robotic connection, magnetic soft-robot components, and magnetorheological fluid control (5–8). In parallel, physical-interface work has shown that embedded magnets can create legible, passive haptic states in digitally fabricated mechanisms (9).

FluxCell connects those threads to a Sarrus-cell robotic-matter concept. The design does not ask a small magnet to pull a compliant mechanism across a large open gap. Instead, a fixed electropermanent magnet cartridge creates a localized magnetic circuit, while steel fork and tongue keepers slide through that circuit as the Sarrus-compatible frame changes width. The first experiment is therefore reduced to a compact overlap problem: pulse the cartridge, record whether geometry changes, check whether the state persists at zero current, reverse the pulse, and measure the limits. This is a stronger scientific object than a broad concept sketch because it can fail cleanly. Either pulse-programmed keeper overlap produces a visible held mechanical state, or the measurement identifies the dominant failure mode.

2 Results

FluxCell is presently a pre-measurement design result. The result is therefore not an efficiency claim or a force-density claim. It is the conversion of a speculative robotic-matter idea into a physically ordered architecture, a locked bill of materials, and a small proof whose required traces are defined before the bench test begins.

2.1 A fixed cartridge turns actuation into keeper overlap

The locked architecture separates the magnetic cartridge from the moving carriers (Fig. 1). The cartridge contains the Alnico pulse element, the NdFeB bias element, and the top and bottom steel pole plates. The moving structure contributes steel keepers only: two fork prongs from one side and a center tongue from the opposite side. The useful magnetic path is therefore local to the overlap between the fixed pole plates and the moving steel keepers. This makes the one-axis proof inspectable because the visible width change and the hidden magnetic variable are linked by one measurable dimension: keeper overlap.

Where the EPM is: fixed center cartridge; prongs/tongues are moving steel keepers

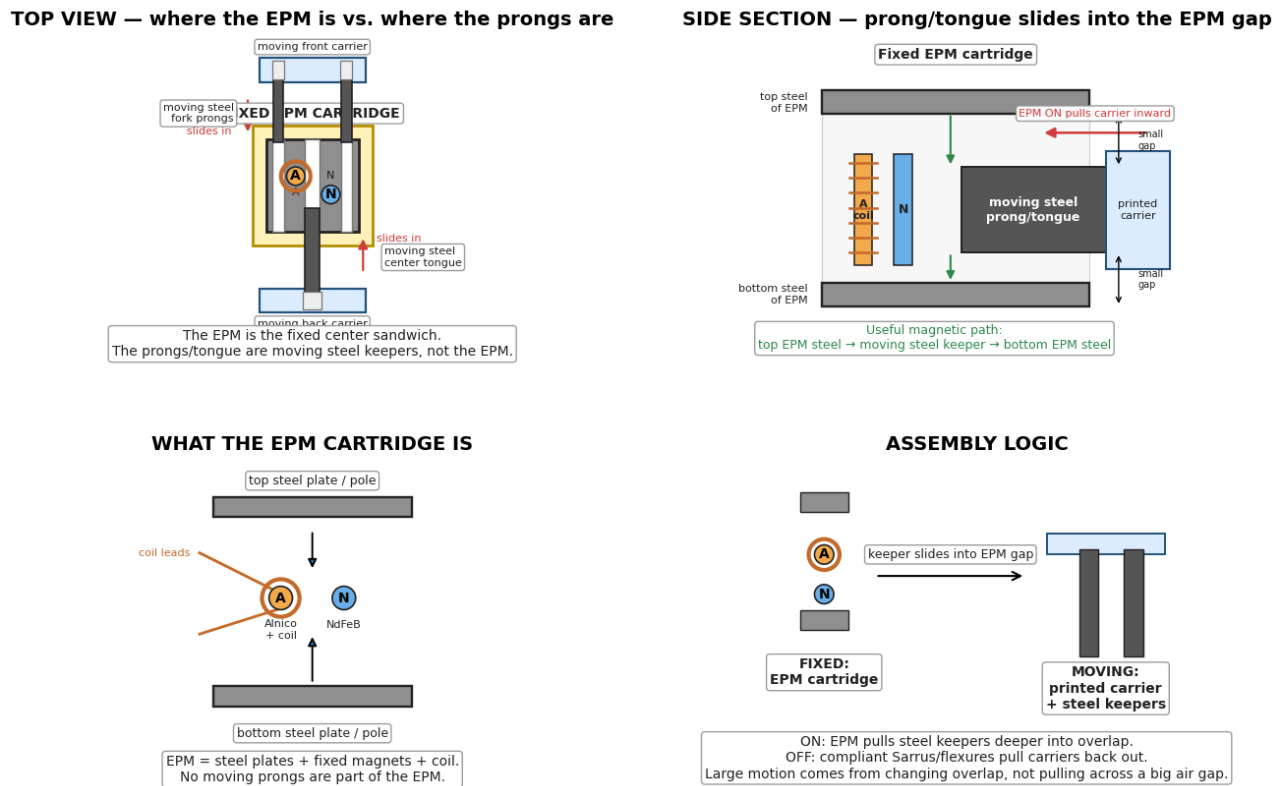


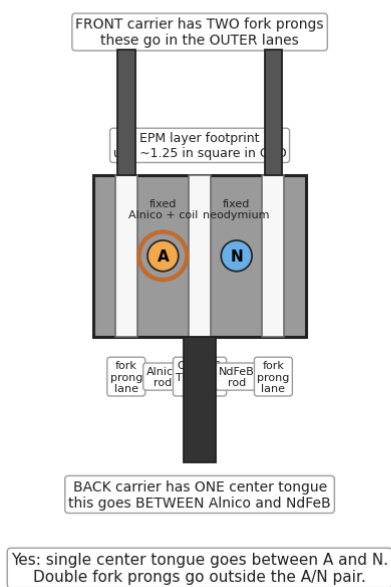
Figure 1. Fixed cartridge and moving keeper concept. The EPM cartridge remains fixed while fork prongs and a center tongue move through the working gap. The useful magnetic path is top pole plate to moving steel keeper to bottom pole plate. Source: FluxCell uploaded design figure, 13 May 2026.

2.2 The lane layout removes collision before fabrication polish

The one-axis layer uses a comb/fork topology (Fig. 2). The center tongue enters between the Alnico and NdFeB rods, while the fork prongs pass outside the pair. The moving steel pieces therefore interdigitate rather than collide. This geometry is a prerequisite for meaningful actuation testing: if the carriers collide, any later force or current measurement would only describe a packaging failure. The lane figure therefore serves as an engineering gate before committing to a monolithic two-axis cell.

Clear lane layout: center tongue between Alnico/NdFeB, fork prongs outside

ONE AXIS LAYER — LANE ORDER



ASSEMBLED TOP VIEW — WIDE/OFF vs PULLED-IN

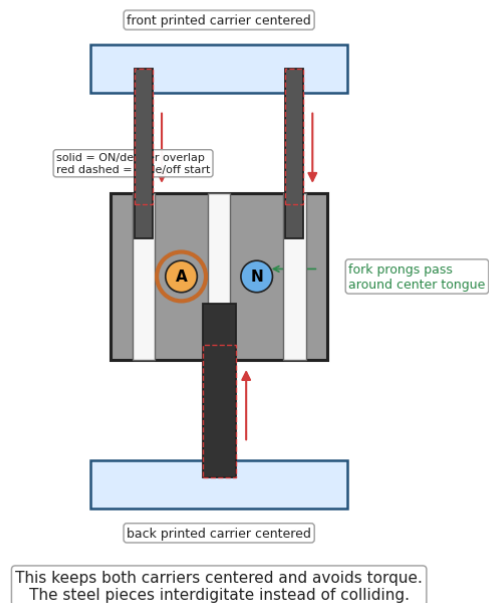
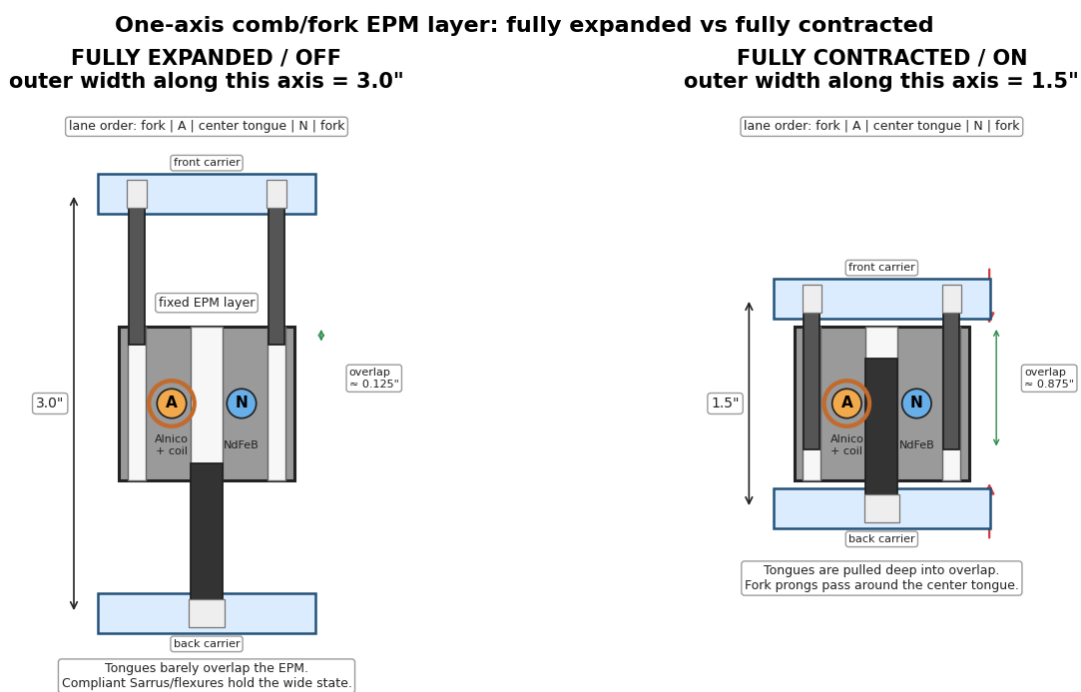


Figure 2. Comb/fork lane order. The center tongue runs between the Alnico and NdFeB positions, and fork prongs run on the outside lanes. The layout keeps both carriers centered and gives the magnetic circuit a repeatable overlap variable. Source: FluxCell uploaded design figure, 13 May 2026.

2.3 The dimensional lock creates a falsifiable mechanical-memory proof

The current dimensional correction uses 1.00 inch keeper inserts rather than 1-1/8 inch inserts. With 1.25 inch square pole plates, the one-axis proof targets a 3.0 inch expanded/off width and a 1.5 inch contracted/on width, corresponding to a 0.75 inch side stroke. The off-state overlap is approximately 0.125 inch and the on-state overlap is approximately 0.875 inch (Fig. 3). These numbers are design targets, not measured performance. Their value is that they make the first physical test falsifiable: a synchronized record can show width, pulse polarity, current, hold at zero drive, release after reverse pulse, and coil temperature.



This shows one axis only. The second axis is the same layer rotated 90° and stacked at a different height.

Figure 3. One-axis overlap target. The expanded/off and contracted/on states create a large visible width change while the magnetic variable stays local to keeper overlap. Current evidence locks the geometry; measured force, current, heat, repeatability, and release margins remain pending until the proof is built. Source: FluxCell uploaded design figure, 13 May 2026.

2.4 The publishable evidence gate is a synchronized trace

The manuscript becomes an actuator paper only if the one-axis proof produces a synchronized record rather than a single success photograph. The minimum publishable trace is deliberately small (Table 1). It has to show the command, the electrical cost of the command, the mechanical response, and whether the state remains after drive current returns to zero. This makes the evidence portable: the same trace can support a robotics claim, a mechanical-memory claim, and a later four-cell surface demonstration.

Table 1. Evidence gate for converting the present design manuscript into an experimental actuator manuscript.

Signal	Required record	Current state
Stroke	Width before pulse, width after pulse, and held width at zero current	pending measurement
Switching cost	Pulse polarity, duration, current limit, and supply voltage	pending measurement
Hold	Zero-current interval with no visible return drift	pending measurement
Release	Opposite pulse followed by measurable release or reverse motion	pending measurement
Thermal margin	Coil temperature note after repeated pulses	pending measurement
Failure mode	If motion fails, dominant limit: friction, weak flux, twist, heat, or residual sticking	pending measurement

2.5 The build lock is small enough to purchase and test

Table 2. Locked purchase logic for one proof plus four complete two-axis cells. Costs are purchase-planning values before shipping and tax, not experimental performance data.

Part	Qty	Locked specification	Approx total
Alnico 5 rods	12	0.250 inch diameter by 0.625 inch long, axial	\$23.88
NdFeB rods	12	K&J D4A, 1/4 inch diameter by 5/8 inch long, N42 axial	~\$15.00
Square pole plates	16	1018 steel, 0.25 inch thick by 1.25 inch wide, cut length 1.25 inch	~\$56.00
Fork prong inserts	24	A36 flat bar, 1/8 inch by 1/2 inch, cut length 1.00 inch	\$167.28
Center tongue inserts	12	1018 flat bar, 1/4 inch by 1/2 inch, cut length 1.00 inch	\$87.72
Wire, tape, supply, leads, pigtails, heat shrink	set	30 AWG magnet wire, Kapton, 0–30 V / 0–10 A supply, leads, silicone wire, strain relief	~\$106.75

The total before shipping and tax is approximately \$456.65 (10–13). This number is not a claim about manufacturability at scale. It is a friction claim: the proof has a specific bill of materials and no longer depends on an undefined part-search step.

3 Discussion

The strongest version of FluxCell is not a generic magnetic actuator paper. It is a paper about pulse-programmed mechanical memory. The state variable is not merely magnetic polarity; it is a visible mechanical configuration coupled to magnetic overlap and held after the electrical pulse ends. If the one-axis proof records clean switching, zero-current hold, reverse-pulse release, bounded heating, and repeated cycles, the work can move from design protocol to actuator evidence.

The source boundary matters because overclaiming would make the project weaker, not stronger. This manuscript does not claim force density, efficiency, safety, lifetime, or reliable two-axis behavior. It claims that the current design has become concrete enough to test and that the proof has been reduced to a small set of measurements. That reduction is the paper's present contribution. It makes the next experiment difficult to hand-wave: either the geometry moves and holds under a measured pulse, or the data identify the dominant failure mode.

For R&D in physical experiences, the compelling artifact would be a small surface that changes contour, holds while electrically quiet, and returns on command. That future demonstrator would be human-readable because the motion, latch, and reset are visible. This is the portfolio-facing reason the project matters for an Imagineering-style R&D path: it combines physical computing, visible mechanism design, soft-robotic structure, rapid fabrication, and evidence-rich iteration in one inspectable artifact. The current paper does not claim that demonstrator exists, nor does it imply any hiring outcome or institutional relationship. It defines the cell-level experiment that must pass before that story is credible.

4 Materials and Methods

4.1 One-axis proof geometry

The proof uses a fixed center EPM cartridge with two 1.25 inch square pole plates made from 0.25 inch by 1.25 inch 1018 steel rectangle bar cut to 1.25 inch length. The magnetic elements are one 0.250 inch diameter by 0.625 inch long Alnico 5 rod and one 1/4 inch diameter by 5/8 inch long NdFeB rod. The moving keepers are two A36 fork prong inserts, each 1/8 inch by 1/2 inch by 1.00 inch, and one 1018 center tongue insert, 1/4 inch by 1/2 inch by 1.00 inch.

4.2 Electrical pulse protocol

The Alnico coil is insulated with Kapton, wound with 30 AWG magnet wire, and connected to an H-bridge driver from a current-limited bench supply. The first test uses short bidirectional pulses separated by zero-drive intervals. For each pulse, the record should include polarity, pulse duration, current limit, observed width, hold state after zero drive, release state after reverse pulse, and a coil temperature note.

4.3 Acceptance boundary

The first proof succeeds only if the one-axis assembly visibly changes width, holds a state after current returns to zero, releases or reverses under the opposite pulse, and stays within a temperature range safe enough for continued bench testing. A partial result remains useful if it identifies the dominant failure mode: weak magnetic force, high friction, carrier twist, thermal limit, or poor release margin.

5 Acknowledgements

This draft was assembled from the FluxCell design log, uploaded schematic figures, and the current locked purchase list.

6 Funding

No external funding is claimed in this draft.

7 Author contributions

Alan N. Pham conceived the project direction, maintained the design log, selected the current purchase lock, and defined the proof target. Codex assisted with manuscript formatting, page integration, figure placement, and verification workflow.

8 Competing interests

The author declares no competing interests for this draft.

9 Data availability

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The current data are design records, uploaded schematic figures, supplier links, and purchase specifications. Measured actuation data are pending the one-axis proof.

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10 Code availability

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The current project page, generated proof bundle, and manuscript source are served through the FluxCell site. Arduino pulse code is included in the generated proof bundle.

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11 Additional information

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This is a draft manuscript and design protocol, not a completed experimental report. The claims should be updated only after the one-axis proof records measured current, width, temperature, hold, release, and repeatability.

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